Time: 60 minutes Max marks: 30

Instructions:
- Do not plagiarize. Do not assist your classmates in plagiarism.
- Define all your variables properly, and show your full solution to get full credit.
- Attempt all questions that you can.
- In the unlikely case a question is not clear, discuss it with an invigilating TA/TF. Please ensure that you clearly include any assumptions you make, even after clarification from the invigilator.

1. (5 points) Answer the following **True/False** questions. A statement is **True** only if it is *always* true; it is **False** if there is even one setting in which it fails. You must justify your answer (a short argument, derivation or counterexample) to get full credit.

   (a) Each iteration of k-means never increases the k-means objective $J = \sum_i \|\boldsymbol{x}_i - \boldsymbol{\mu}_{c(i)}\|^2$, where $\boldsymbol{\mu}_{c(i)}$ are the nearest cluster center for point $\boldsymbol{x}_i$.

   Solution: **True.** A k-means iteration has two steps, and neither can increase J.
   Assignment: each point is assigned to its *nearest* centre, so each term $\|\boldsymbol{x}_i - \boldsymbol{\mu}_{c(i)}\|^2$ can only stay the same or decrease.
   Update: For the kth cluster, $J_k = \sum_{i \in C_k} \|\boldsymbol{x}_i - \boldsymbol{\mu}\|^2$ is minimised at $\boldsymbol{\mu} = \sum_{i \in C_k} \boldsymbol{x}_i$ the cluster mean (set its gradient $\nabla J_k = -2 \sum_{i \in C_k} (\boldsymbol{x}_i - \boldsymbol{\mu}) = 0$), so replacing each centre by its mean can only decrease J. Even for empty clusters, their respective centres are left unchanged and J still does not increase.).

   (b) For i.i.d. samples $x_1, \ldots, x_n$ ($n \geq 2$) from a normal distribution, $\mathcal{N}(\mu, \sigma^2)$ with both μ and σ^2 unknown, the unbiased estimate of the variance is given by $s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$. This s^2 is also the maximum likelihood estimate of σ^2.

   Solution: **False.** The log-likelihood is $\log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2$. Setting $\partial / \partial \mu = 0$ gives $\hat{\mu} = \bar{x}$; then $\partial / \partial \sigma^2 = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (x_i - \bar{x})^2 = 0$ gives

   $$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2 = \frac{n-1}{n} s^2 \neq s^2.$$

   In this case, the MLE is biased, $\mathbb{E}[\hat{\sigma}^2_{\text{MLE}}] = \frac{n-1}{n} \sigma^2$; the $\frac{1}{n-1}$ correction that makes s^2 unbiased is not what maximises the likelihood.

   (c) A coin shows 0 heads in 5 tosses. The maximum likelihood estimate of the heads probability $p \in [0, 1]$ is 0.

   Solution: **True.** $L(p) = p^0 (1-p)^5 = (1-p)^5$, which is strictly decreasing on $[0, 1]$, so it is maximised at $p = 0$ (the general MLE k/n with $k = 0$). However, note that setting $dL/dp = 0$ does not find it: the maximum is at the boundary of the parameter space. (This "zero-count" estimate is exactly what a prior, i.e. MAP estimation, avoids.) (Or, Full credit only when wrote MLE expression and argued that as number of heads is zero the estimate would be zero.)

   (d) Consider logistic regression with an intercept term, i.e., the b in $(\boldsymbol{w}^T \boldsymbol{x} + b)$. When fit using unregularized maximum likelihood, the average predicted probability on the training set equals the fraction of positive training labels.

   Solution: **True.** With $p_i = \sigma(\boldsymbol{w}^\top \boldsymbol{x}_i + b)$, the log-likelihood $\sum_i \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right]$ is differentiable and the MLE is an unconstrained maximum, so its gradient is zero there. The derivative $\frac{\partial L}{\partial b}$ with respect to the intercept is $\sum_i (y_i - p_i)$, hence

   $$\sum_i (y_i - p_i) = 0 \implies \frac{1}{n} \sum_i p_i = \frac{1}{n} \sum_i y_i.$$

   (e) Since sigmoid is a non-linear function, the decision boundary of logistic regression on the raw input features $\boldsymbol{x}$ (predict 1 iff $P(y = 1 \,|\, \boldsymbol{x}) \geq 0.5$) is non-linear in $\boldsymbol{x}$.

Solution: **False.** $\sigma(t) \geq 0.5 \iff t \geq 0$, so the rule is $\boldsymbol{w}^\top \boldsymbol{x} + b \geq 0$: the decision boundary $\{\boldsymbol{x} : \boldsymbol{w}^\top \boldsymbol{x} + b = 0\}$ is a hyperplane in $\boldsymbol{x}$. The sigmoid function is monotonically increasing, so it changes how confident the prediction is, not where the boundary lies.

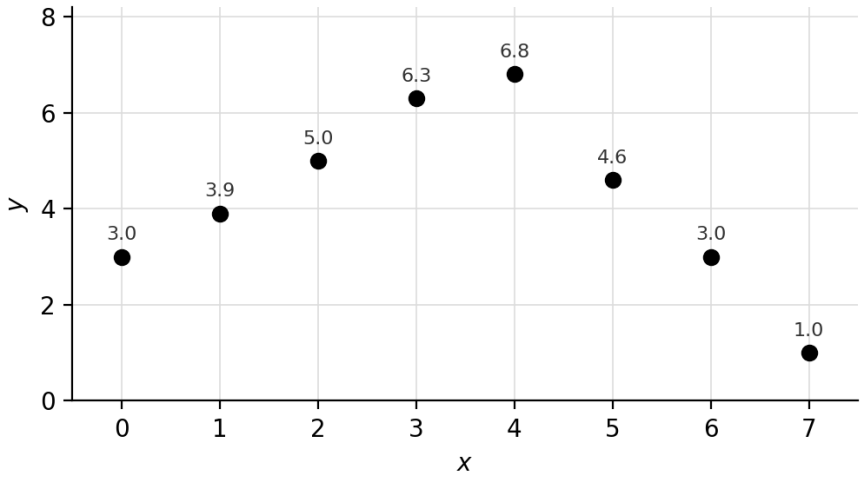


Figure 1: Figure for Question 2

2. (10 points) For the data shown in Fig. 1, we have eight noisy observations $\{(x_i, y_i)\}_{i=1}^{8}$ were evidently generated as $y_i = f(x_i) + \varepsilon_i$. Here $f(x_i)$ is a 2-piece, piecewise linear function.

   (a) (2 points) By visually observing the figure, write the 2-piece, piecewise linear function you would use to model the given data. Give your solution in the following form using appropriate parametrization.

$$\widehat{f}(x) = \begin{cases} \text{parametric form of pc. 1}, & \text{condition for pc. 1} \\ \text{parametric form of pc. 2}, & \text{condition for pc. 2} \end{cases}$$

   Solution: The data rise roughly linearly up to $x \approx 3$–4 and then fall roughly linearly, so

$$\widehat{f}(x) = \begin{cases} \widehat{f}_1(x) = a_1 + b_1 x, & x < c \\ \widehat{f}_2(x) = a_2 + b_2 x, & x \geq c \end{cases}$$

   with parameters a_1, b_1, a_2, b_2 and breakpoint c.
   From the figure: $b_1 > 0$, $b_2 < 0$ and c between 3 and 4; e.g.

$$\widehat{f}(x) = \begin{cases} 3 + x, & x < 3.5 \\ 14 - 1.9x, & x \geq 3.5 \end{cases}$$

   [1 mark: two linear pieces with the correct parametrization; 1 mark: an explicit breakpoint condition. If explicit numbers are given, check if the sign of the slopes are correct; if they are incorrect, a small penalty should be applied (-0.5).]; no penalty for not reporting explicit numbers

   (b) (2 points) Using only logistic regression and linear regression models, write an algorithm inspired by the k-means clustering method, to estimate the necessary parameters of your model for $\widehat{f}(x)$.

   Solution: Treat the two lines like k-means centroids ($k = 2$), and the absolute residual like the distance to a centroid, and use a logistic regression model as a "gate" to decide the piece from x alone. The loss function would then be
   Let $z_i \in \{0, 1\}$ be the piece label of point i (0: piece 1, 1: piece 2).

$$J = \sum_{i=1}^{8} \left(z_i \left(y_i - a_1 - b_1 x_i \right)^2 + (1 - z_i) \left(y_i - a_2 - b_2 x_i \right)^2 \right)$$

1. **Initialise** the labels, e.g. $z_i = \mathbb{1}[x_i \geq c_0]$ for a guessed breakpoint c_0 (or start from two guessed lines and do step 3 first).
2. **Fit the experts (update step).** Fit $\widehat{f}_1(x) = a_1 + b_1 x$ by ordinary least squares to the points with $z_i = 0$, and $\widehat{f}_2(x) = a_2 + b_2 x$ to those with $z_i = 1$.
3. **Re-assign (assignment step).** $z_i = 1$ if $\Big(y_i - \widehat{f}_2(x_i)\Big)^2 < \Big(y_i - \widehat{f}_1(x_i)\Big)^2$, else $z_i = 0$.
4. **Adjust the gating function.** Fit logistic regression $P(z = 1 \mid x) = \sigma(wx + w_0)$ to the pairs (x_i, z_i), and relabel $z_i = \mathbb{1}[\sigma(wx_i + w_0) \geq 0.5]$. In 1-D this is a single threshold, so each piece becomes an interval of x, with breakpoint $\hat{c} = -w_0/w$.
5. **Repeat** steps 2–4 until the labels stop changing.

Prediction for a new x (no y available): choose the piece with the gate, $z = \mathbb{1}[\sigma(wx+w_0) \geq 0.5]$, then return $\widehat{f}_1(x)$ or $\widehat{f}_2(x)$.

[1 mark: alternating "assign by residual / refit lines by least squares" loop; 1 mark: logistic regression used to map x to a piece (threshold $\hat{c} = -w_0/w$).]

(c) (6 points) Write the likelihood function as $L(\theta) = \prod_{i=1}^{8} P(y_i \mid x_i)$, where you also need to define the set of learnable parameters $\theta = \{\cdot\}$. These are the parameters for the piecewise linear model you previously developed using linear and logistic regression models. Does a closed form solution exist for this likelihood function? Explain why or why not. {*Hint*: Introduce a binary latent variable z, and use it to write $P(y \mid x)$ leveraging the two linear regression models that you used for modeling $f(x)$.}

Solution: **The Model.** Let $z_i \in \{0, 1\}$ be the latent variable deciding the piece of point i. We model the prediction of this *unobserved* variable using the logistic regression model, i.e., $P(z_i = 1 \mid x_i) = \pi_i = \sigma(wx_i + w_0)$, $P(z_i = 0 \mid x_i) = 1 - \pi_i$. The other two linear functions $\widehat{f}_1(x)$ and $\widehat{f}_2(x)$ are modeled using linear regression with learnable parameters (a_1, b_1), (a_2, b_2) and Gaussian noise, $\varepsilon \in \mathcal{N}(0, s^2)$.

- Logistic regression: $P(z_i = 1 \mid x_i) = \pi_i = \sigma(wx_i + w_0)$, $P(z_i = 0 \mid x_i) = 1 - \pi_i$.
- Linear Regression models with Gaussian noise:

$$p(y_i \mid x_i, z_i = 0) = \mathcal{N}\big(y_i;\ \mu = a_1 + b_1 x_i,\ \sigma^2 = s^2\big),$$
$$p(y_i \mid x_i, z_i = 1) = \mathcal{N}\big(y_i;\ \mu = a_2 + b_2 x_i,\ \sigma^2 = s^2\big),$$

where $\mathcal{N}(y; \mu, s^2) = \frac{1}{\sqrt{2\pi s^2}} \exp\big(-\frac{(y-\mu)^2}{2s^2}\big)$ and s^2 is the noise variance. Write $\mathcal{N}_1(y_i)$ and $\mathcal{N}_2(y_i)$ for these two densities.

Learnable Parameters: $\theta = \{w,\ w_0,\ a_1,\ b_1,\ a_2,\ b_2,\ s^2\}$ (separate variances s_1^2, s_2^2 are also acceptable).

Likelihood. Marginalise the latent variable, $p(y_i \mid x_i) = \sum_{z \in \{0,1\}} P(z \mid x_i)\, p(y_i \mid x_i, z)$:

$$L(\theta) = \prod_{i=1}^{8} \Big[\big(1 - \sigma(wx_i + w_0)\big)\,\mathcal{N}\big(y_i;\ a_1 + b_1 x_i,\ s^2\big) + \sigma(wx_i + w_0)\,\mathcal{N}\big(y_i;\ a_2 + b_2 x_i,\ s^2\big)\Big].$$

(This is a *mixture of experts*: the probabilities from the "gate" weight the two linear-regression likelihoods.)

No closed-form solution.

First, $\log L(\theta) = \sum_i \log\big[(1 - \pi_i)\mathcal{N}_1(y_i) + \pi_i \mathcal{N}_2(y_i)\big]$ has a *sum inside the* log. Unlike linear regression, it does not simplify into separate terms, so setting $\nabla_\theta \log L = 0$ gives coupled non-linear equations. So θ must be found iteratively. Secondly, even if the z_i were known, the gating part is achieved via a logistic regression model, whose MLE has no closed form (it needs iterative optimisation).

With z_i known, the complete-data log-likelihood separates:

$$\log L_c(\theta) = \underbrace{\sum_i \left[z_i \log \pi_i + (1 - z_i) \log(1 - \pi_i)\right]}_{\text{logistic regression in } (w, w_0)} + \underbrace{\sum_i \left[(1 - z_i) \log \mathcal{N}_1(y_i) + z_i \log \mathcal{N}_2(y_i)\right]}_{\text{least squares on each piece}},$$

which is why the algorithm in part (b) — alternately assigning z and fitting the lines separately — is akin to the k-means algorithm.

[1 mark: θ listed correctly; 0.25 deducted if s^2 is not mentioned (no penalty if $w = 1$ is assumed); 1 mark: gating using logistic regression $P(z \mid x) = \sigma(wx+w_0)$; 1 mark: the two Gaussian linear-regression likelihoods; 1 mark: correct marginal $L(\theta)$ (sum over z inside the product); 2 marks: "no closed form" with a valid reason — sum inside the log / coupled non-linear equations (1), and logistic MLE has no closed form (1).]

3. (5 points) Consider a binary classification problem where the class label is given by $Y \in \{0, 1\}$ with prior probabilities $P(Y = 1) = \phi$ and $P(Y = 0) = 1 - \phi$. The input feature vector is continuous and d-dimensional, $\mathbf{x} = [x_1, x_2, \ldots, x_d]^T \in \mathbb{R}^d$. Suppose we model the class-conditional distributions using Gaussian distributions.

   (a) (3 points) Describe binary classifiers that you would design for this problem using (i) Gaussian Naive Bayes (GNB), where the conditional probabilities are modeled using a Gaussian distribution and (ii) Gaussian Discriminant Analysis (GDA). Define all variables.

   Solution:

   Let the training data be $\{(\boldsymbol{x}^{(j)}, y^{(j)})\}_{j=1}^n$ and $n_k = \sum_j \mathbb{1}[y^{(j)} = k]$ be the number of examples of class k. As both the classifiers, GNB and GDA are generative, fit $P(Y)$ and $p(\boldsymbol{x} \mid Y)$ by maximum likelihood. Then the following decision rule uses the posterior probability $P(Y \mid \boldsymbol{x})$, which in turn is computed by applying Bayes' rule.

$$\hat{y}(\boldsymbol{x}) = 1 \iff \log \frac{P(Y = 1 \mid \boldsymbol{x})}{P(Y = 0 \mid \boldsymbol{x})} = \log \frac{\phi}{1 - \phi} + \log \frac{p(\boldsymbol{x} \mid Y = 1)}{p(\boldsymbol{x} \mid Y = 0)} \geq 0.$$

   Or equivalently

$$\hat{y}(\boldsymbol{x}) = 1 \iff P(Y = 1 \mid \boldsymbol{x}) \geq P(Y = 0 \mid \boldsymbol{x}).$$

   In both the classifiers, $\hat{\phi} = n_1/n$.

   - In a continuous **Gaussian Naive Bayes (GNB)** classifier, we assume features are conditionally independent given the class label, meaning $P(X_i|Y = k) \sim \mathcal{N}(\mu_{ki}, \sigma_{ki}^2)$ for each feature $i \in \{1, \ldots, d\}$ and each class $k \in 0, 1$. Therefore the joint class conditional distribution is given as $P(\mathbf{X} \mid Y = k) \sim \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k^{GNB})$, where Σ_k^{GNB} is a diagonal covariance matrix with entries as σ_{ki}^2 and $\boldsymbol{\mu}_k$ is the mean for class k.
   - In **Gaussian Discriminant Analysis (GDA)**, we assume $P(\mathbf{X}|Y = k) \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k^{GDA})$, Σ_k^{GDA} is the covariance matrix and $\boldsymbol{\mu}_k$ is the mean for class k.

   No penalty for assuming same variance of each class, or even for each feature x_i.

   (b) (2 points) Formally state the condition(s) such that the general Gaussian Discriminant Analysis classifier matches the Gaussian Naive Bayes classifier exactly.

   Solution:

   (c) For the Gaussian Discriminant Analysis (GDA) framework to match the continuous Gaussian Naive Bayes (GNB) assumptions exactly, two distinct constraints must be applied to the covariance matrix $\boldsymbol{\Sigma}$:

   1. **Diagonal Structure:** The shared covariance matrix $\boldsymbol{\Sigma}$ must be strictly diagonal to represent total conditional independence between individual features given the class label (i.e., all off-diagonal elements must be zero).

2. **Feature-Specific Variances:** The elements along the principal diagonal must correspond exactly to the unique, feature-specific conditional variances σ_i^2.

Mathematically, the matrix must take the exact form:

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_d^2 \end{bmatrix} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_d^2)$$

1 mark: Explicitly stating that the covariance matrix $\mathbf{\Sigma}$ must be restricted to a diagonal format (or stating that all off-diagonal entries $\Sigma_{ij} = 0$ for $i \neq j$).
1 Mark: Explicitly mapping the diagonal elements to the individual feature variances σ_i^2.

4. (10 points) A state power-distribution company wants to flag households that are stealing electricity. Theft is confirmed only when an inspector visits the house and historically about **1.5%** households were identified with tampered meters.

An intern trained a **Regularized logistic regression** classifier with an ℓ_2-norm regularizer, using 24 months of smart-meter readings (Jan. 2024 – Dec. 2025) from 40,000 meters. Each row is one meter-month with several features, labelled *theft* if an inspection confirmed tampering that month. The intern's report states:

1. I shuffled all 960,000 rows and split them 80% train / 20% test.
2. I tried Regularization parameter $\lambda \in \{10^{-3}, 10^{-2}, \dots, 10^{3}\}$ and kept the one with the highest test accuracy.
3. Training accuracy 99.3%, test accuracy 99.1%. The two are close, so the model neither underfits nor overfits.

Pilot (Jan. 2026). The model was run on one sub-division of 10,000 meters. As a one-time exercise, *every* meter there was inspected: 150 thefts were found. The model had flagged 400 meters for theft, of which 111 were confirmed thefts.

(a) (3 points) The chief engineer writes: *"The intern promised 99.1%. The pilot gives 96.7% — worse than just calling every household honest. Should we scrap the model?"* Give your opinion on the model's capability in at most four sentences, quoting whatever numbers / metrics from the pilot to support your recommendation.

Solution: Pilot confusion matrix: TP = 111, FP = 400 − 111 = 289, FN = 150 − 111 = 39, TN = 10000 − 150 − 289 = 9561. Accuracy = (111 + 9561)/10000 = 96.7%; declaring everyone honest gives (10000 − 150)/10000 = 9850/10000 = 98.5%.

Response for using / scrapping the Model : The model *should not be scrapped* because *accuracy* is an inappropriate metric in this case as the data is severely imbalanced with only 1.5% of households being detected as theft cases. The "everyone is honest" rule scores a 98.5% accuracy but catches nobody (recall 0).

The intern's model caught 111 of the 150 thieves (recall = 111/150 ≈ 0.74), and 111 of its 400 flags were real (precision = 111/400 ≈ 0.28). Randomly inspecting 400 households would have found only about 6 thefts (precision = 0.015), so the model is about 0.28/0.015 ≈ 18× better at directing inspectors than randomly inspecting households.

[1 mark: accuracy vs. the trivial baseline and why it misleads under imbalance; 1 mark: correct precision and recall; 1 mark: a justified keep/do-not-scrap conclusion.]

(b) (3 points) The intern says the pilot was just unlucky. For each of the three statements in the intern's report, say if it contributed to 99.1% being an unreliable promise, and why.

Solution: All three lines contributed. *[1 mark each]*

- **Line 1 (shuffled row split).** Adjacent months of the *same* meter land in both train and test, so the model is tested on near-copies of training rows, and future months help predict past ones. The deployed model scores *future* months, so the split must be across time: train on earlier months, test on later months (splitting by meter is also acceptable).

- **Line 2 (choosing λ by test accuracy).** The test set was used to pick the model, so the reported test score is overly optimistic, whereas the reported test score should be an unseen-data estimate. The λ must be chosen on validation data (a separate validation split or cross-validation), leaving the test set untouched.
- **Line 3 ("close accuracies $\Rightarrow$ good fit").** Since the data is severely imbalanced, using accuracy as a metric for comparing train and test set performance is a bad choice, and the train-test gap does not adequately capture the model's overfitting / underfitting performance, and instead should be compared in terms of precision and recall or F1-score on the training vs. validation data.

(c) (4 points) For the next version, the team compares three prior regularzation parameters (λ) using 5-fold cross-validation (CV) on the 2024–25 data.

Reg. param. λ	Train precision	Train recall	CV precision	CV recall
10	0.30	0.54	0.27	0.56
1	0.45	0.80	0.32	0.66
0.1	0.90	0.97	0.22	0.71

The inspectors' union will not accept more than 3 wasted visits out of every 4. The board wants as many thieves caught as possible.

i. (3 points) Which λ should be deployed? Explain your choice over each of the others.

ii. (1 point) The team plans to tell the chief engineer: *"Expect about 1 in 3 visits to find theft, and about two-thirds of all thefts to be caught."* Is that a reliable claim? If not, what should the team do before quoting any number?

Solution: (i) The union's condition is at least one theft (True Positive: TP) out of 4 inspections (Total Positive: TP + FP), i.e., **precision** ≥ 0.25. The board's wish is **maximise recall** subject to the union's condition. Therefore, we are looking for a model that has precision of at least 0.25 and the highest recall. However, the decision should be made on the cross-validation data, and not the training data (where the performance will be significantly more optimistic, and we will get fooled by an overfitted model).

- $\lambda = 10$: Too much regularization. The model somewhat **underfits** (train and CV both low and nearly equal), even though the precision constraint is met, the recall is still low.
- $\lambda = 0.1$: Regularization is too weak. The model is closer to an MLE fit and tends to **overfit** (train precision 0.90 vs. CV 0.22). Additionally, it has the highest CV recall but fails the precision constraint.
- **Deploy** $\lambda = 1$: CV precision $0.32 \geq 0.25$ with the highest recall (0.66) among the values that satisfy it. The train-val performance gap is moderately small. The gap suggests that a better model can be trained with other interventions, however, among the ones already trained, the choice is made on the CV scores.

[1 mark: correct choice via precision ≥ 0.25 then max recall on CV numbers.
1 mark: correct under/over-fitting diagnosis of 0.1 and 10.
Choosing 10 because of its training scores, or because of CV recall alone, earns 0 for the first mark.]

(ii) **Not reliable.** The precision of 0.32 and corresponding recall of 0.66 are the *best* of the three CV results on the same data that was used to choose λ, so they are still an optimistic estimate of the test performance. Before making this claim, the team should hold out the most recent months as a **test set that is never used for selection**, i.e., form a three-way split (train / validation-or-CV / test), choose λ by CV on the earlier data, refit on all of it, and evaluate *once* on the test months and quote *those* precision and recall numbers. Alternatively, evaluating the refitted model once on the January-2026 Pilot data, which played no part in selection of λ, is equally acceptable.

[1/2 mark: why the CV figure is biased.
1/2 mark: a correct protocol.]